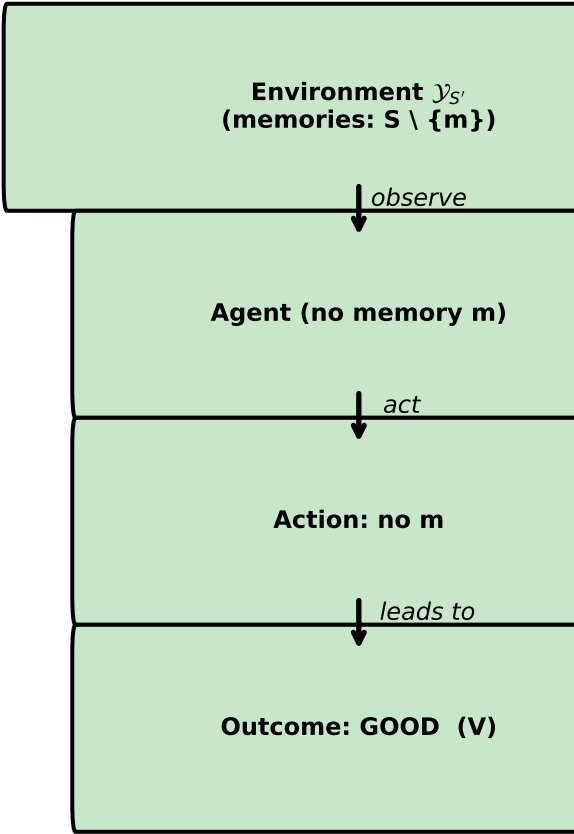
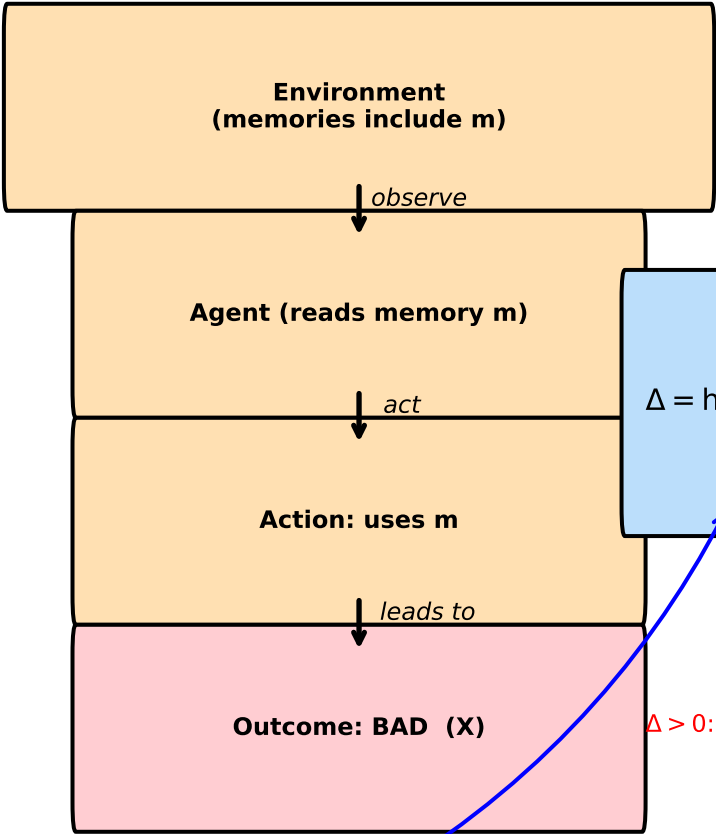


$\mathcal{Y}_S$ (with memory m)

$\mathcal{Y}_{S'}$ (without memory m)

Treatment
(memory m present)

Control
(memory m deleted)



$$\Delta = \text{harm}(\mathcal{Y}_S) - \text{harm}(\mathcal{Y}_{S'})$$

$\Delta > 0$: memory m causes harm

$$\text{harm}(\mathcal{Y}_S) = 1$$

$$\text{harm}(\mathcal{Y}_{S'}) = 0$$

Definition: $\text{harm}(\mathcal{Y}_S) = \mathbb{1}[\text{action is bad in } \mathcal{Y}_S]$; $\Delta = \text{harm}(\mathcal{Y}_S) - \text{harm}(\mathcal{Y}_{S'})$ measures causal effect of memory m on harm.